CS5234 Quiz 1 · Cheat Sheet

L1–L5 + Ex. 1–3 + HW1 · log = ln · w.h.p. = $1-1/\text{poly}(n)$

0 · Recognise the problem → tool

- **Upper bound (design + analyse)**: write the target as a sum of $[0,1]$ (or bounded) variables → unbiased estimator → cost per sample × #samples → concentration picks #samples
- **Lower bound $\Omega(\cdot)$**: Yao. Two hard distributions with different answers; show q queries give (almost) the same transcript
- **Property testing**: structural lemma "ε-far ⇒ many witnesses (bad indices / pairs / edges)" + "sampling finds one"
- **Streaming**: keep a small summary; prove invariant by induction over the stream

Exam answer skeleton (1) Define r.v.s explicitly. (2) Compute $\mathbb{E}$. (3) Bound Var or state independence. (4) Name the inequality + plug numbers. (5) State final success prob. & query/space cost.

1 · Probability basics

- Linearity: $\mathbb{E}[\sum X_i]=\sum\mathbb{E}[X_i]$ (no independence needed); $\mathbb{E}[\mathbf{1}_A]=\Pr(A)$
- $\text{Var}X=\mathbb{E}[X^2]-(\mathbb{E}X)^2=\mathbb{E}[(X-\mathbb{E}X)^2]$; $\text{Var}(aX+b)=a^2\text{Var}X$
- $X\in\{0,1\}$: $\text{Var}X=p(1-p)\le p=\mathbb{E}X$; $X\in[0,M]$: $\mathbb{E}X^2\le M\,\mathbb{E}X$
- Independent: $\mathbb{E}[XY]=\mathbb{E}X\mathbb{E}Y$, $\text{Var}(X+Y)=\text{Var}X+\text{Var}Y$
- $\text{Var}(\sum_i X_i)=\sum_i\text{Var}X_i+2\sum_{i<j}\text{Cov}(X_i,X_j)$, $\text{Cov}(X,Y)=\mathbb{E}[XY]-\mathbb{E}X\mathbb{E}Y$
- Pairwise indep. ⇒ Cov $=0$; neg. correlated ($\mathbb{E}[X_iX_j]\le\mathbb{E}X_i\mathbb{E}X_j$) ⇒ $\text{Var}(\sum)\le\sum\text{Var}$
- Total expectation: $\mathbb{E}X=\sum_y\Pr(Y{=}y)\mathbb{E}[X|Y{=}y]$ (Morris, recursions)
- Disjoint events are *not* independent (unless prob. 0)
- Distributions: $\text{Bin}(k,p)$: $\mathbb{E}=kp$, Var $=kp(1-p)$; Geometric: $\mathbb{E}=1/p$; Uniform sample of s from n w/o repl.: $\Pr[i\in S]=\frac{s}{n}$, $\Pr[i,j\in S]=\frac{s(s-1)}{n(n-1)}$

2 · Concentration inequalities

Markov $X\ge0$: $\Pr(X\ge t)\le\mathbb{E}X/t$; $\Pr(X\ge\alpha\mathbb{E}X)\le1/\alpha$.
Pf: $\mathbb{E}X\ge t\Pr(X\ge t)$. Use: running time, "few bad high-degree vertices", $\Pr[X\ge1]\le\mathbb{E}X$ for counts

Chebyshev $\Pr(|X-\mathbb{E}X|\ge t)\le\text{Var}X/t^2$.
Relative: $\Pr(|X-\mathbb{E}X|\ge\varepsilon\mathbb{E}X)\le\frac{\text{Var}X}{\varepsilon^2(\mathbb{E}X)^2}$.
Average of k i.i.d. copies: $\Pr(|\bar X-\mu|\ge\varepsilon\mu)\le\frac{\text{Var}X}{k\varepsilon^2\mu^2}$ ⇒ $k=\frac{3\text{Var}X}{\varepsilon^2\mu^2}$ **gives** $2/3$
Only needs pairwise independence; with neg. correlation Var $\le\mathbb{E}$ ⇒ $\Pr(X=0)\le\Pr(|X-\mu|\ge\mu)\le\frac{1}{\mu}$

Chernoff indep., $X_i\in[0,1]$, $X=\sum X_i$, $\mu=\mathbb{E}X$:

$$\Pr(|X-\mu|\ge\delta\mu)\le2e^{-\delta^2\mu/3}\quad(0<\delta\le1)$$

$$\Pr(X\ge(1+\delta)\mu)\le e^{-\delta\mu/3}\quad(\delta>1)$$

$$\Pr(X\ge(1+\delta)\mu)\le\left(\frac{e^\delta}{(1+\delta)^{1+\delta}}\right)^\mu$$

$$\Pr(X\le(1-\delta)\mu)\le\left(\frac{e^{-\delta}}{(1-\delta)^{1-\delta}}\right)^\mu$$

! Upper tail stays valid if μ is replaced by any upper bound $\mu_H\ge\mu$ (use when μ unknown/small)
Sample sizes: relative error ε for $\Pr[\text{hit}]=p$: $k=\frac{3\ln(2/\delta)}{\varepsilon^2p}$; additive $\pm\varepsilon$ on a fraction: $k=O(\log(1/\delta)/\varepsilon^2)$

Hoeffding indep. $X_i\in[0,1]$: $\Pr(|\bar X-\mathbb{E}\bar X|\ge\varepsilon)\le2e^{-2k\varepsilon^2}$ ⇒ $k=\frac{\ln(2/\delta)}{2\varepsilon^2}$

Union bound $\Pr(\bigcup A_i)\le\sum\Pr(A_i)$, no independence. Typical: n events each $\le n^{-c}$ ⇒ $\le n^{1-c}$; $2^{|C|}$ partitions each $e^{-\Theta(|C|)}$

Which one? only $\mathbb{E}$ → Markov · variance known / pairwise indep. / dependent indicators → Chebyshev · fully indep. bounded → Chernoff/Hoeffding (exponential tail; pairs well with union bound)

3 · Amplification & budgets

Mean trick $\mathbb{E}X=A$, $\text{Var}X=M$; average k: $\text{Var}\bar X=M/k$

Median trick single run correct w.p. $\ge2/3$. Median of k is wrong only if $\ge k/2$ runs are wrong on the same side; Y=#runs $>A+\varepsilon$, $\mathbb{E}Y<k/3$; Chernoff: $\Pr(Y\ge k/2)<e^{-k/100}$ ⇒ $\Pr(\text{fail})<2e^{-k/100}$ ⇒ $k=O(\log\frac{1}{\delta})$. Needs success $>1/2$

Standard combo mean of $3M/(\varepsilon^2A^2)$ runs → 2/3; median of $O(\log\frac{1}{\delta})$ such means → $1-\delta$. Total $O\big(\frac{M}{\varepsilon^2A^2}\log\frac{1}{\delta}\big)$

Majority vote decision problem, success $\frac12+\gamma$ → $\Theta(1/\gamma^2)$ runs to reach 2/3

Expected → worst case $\Pr(Q\ge10T)\le0.1$; truncate at $10T$, output anything ⇒ success $\ge\frac23-0.1>\frac12$ ⇒ $O(1)$ repeats + majority ⇒ worst $O(T)$

Budgets error $\varepsilon n=\frac{\varepsilon n}{2}$(bias: truncation / ignored parts)$+\frac{\varepsilon n}{2}$(sampling); failure $\frac13=\sum$ of bad events (union bound); success of "A and B" $\ge1-\Pr\bar A-\Pr\bar B$ or product if independent

4 · Sampling templates (L1-L2, HW1)

Balls & bins n balls: $\mathbb{E}[\text{bin }j]=1$, $\mathbb{E}[\#\text{empty}]=n(1-\frac1n)^n\approx\frac{n}{e}$; empty indicators neg. corr.: $\Pr(j,l\text{ empty})=(1-\frac2n)^n\le(1-\frac1n)^{2n}$. $10n\log n$ balls: bin 2 has $\mu=10\log n$, Chebyshev $\frac{1}{10\delta^2\log n}$ vs Chernoff $2n^{-10\delta^2/3}$; $\Pr(\text{bin empty})\le e^{-10\log n}=n^{-10}$, union ⇒ n^{-9}

Counting 1s, $\pm\varepsilon n$ sample k positions, output $X\frac{n}{k}$; $\mu=\frac{kA}{n}$; need $|X-\mu|\le\varepsilon k$ ⇒ $\delta=\frac{\varepsilon n}{A}$.
$A\ge\varepsilon n$ ($\delta\le1$): $2e^{-\delta^2\mu/3}=2e^{-\varepsilon^2nk/(3A)}\le2e^{-\varepsilon^2k/3}$;
$A<\varepsilon n$ ($\delta>1$): lower side impossible; $e^{-\delta\mu/3}=e^{-\varepsilon k/3}$.
⇒ $k=100/\varepsilon^2$ **gives** ≤0.1**, independent of** n

Median rank in $[(\frac12\pm\varepsilon)n]$: sample k, output sample median. Fail iff $\ge k/2$ samples below $(\frac12-\varepsilon)n$ (or above). Y=#low, $\mathbb{E}Y=(\frac12-\varepsilon)k$, $\frac{k}{2}\ge(1+2\varepsilon)\mathbb{E}Y$ ⇒ Pr $<e^{-\varepsilon^2k/10}$ ($\varepsilon<0.1$). $k=100/\varepsilon^2$, two sides ⇒ 0.8

Hamming distance (HW1 P1) $Z_j=\mathbf{1}[x_{i_j}\ne y_{i_j}]$, $\hat d=\frac{n}{k}\sum Z_j$; Hoeffding $2e^{-2k\varepsilon^2}\le\delta$ ⇒ $k=\frac{\ln(2/\delta)}{2\varepsilon^2}$, $2k$ queries. Alt: Chernoff $O(1/\varepsilon^2)$ + median trick

Sum of values in $[1,M]$ $X=n\,a_I$: $\mathbb{E}X=S$, $\mathbb{E}X^2=n\sum a_i^2\le nMS$ ⇒ $\frac{\text{Var}}{S^2}\le\frac{nM}{S}\le M$ ⇒ $O(M/\varepsilon^2)$

Designing estimators (i) choose what each element contributes so contributions sum to the target (representative / $1/|C|$ / ordered edges); (ii) if variance too big: redistribute credit, split heavy/light, truncate and pay bias

5 · Graph query algorithms

Models adj. matrix: pair query (u,v) (dense); adj. list: degree(v), i-th neighbour(v) (sparse)

Edges $\pm\varepsilon n^2$ sample k pairs, output $X\binom{n}{2}/k$ = counting on $\binom{n}{2}$ bits ⇒ $k=O(1/\varepsilon^2)$; $1-\delta$: $O(\log\frac{1}{\delta}/\varepsilon^2)$

Edges $(1\pm\varepsilon)$ $m=0$ vs 1 needs $\Omega(n)$ (hidden edge) ⇒ assume connected, $m\ge n-1$. Naive d_v sampling: Var $=O(n\bar d)$ (star) ✗.
Order by (degree, id). d'_v=#neighbours ranked above v; $\sum_v d'_v=m$. Estimator: random v, query d_v, random neighbour u, query d_u; $x=d_v$ if $v<u$ else 0. $\mathbb{E}x=\frac1n\sum_v d_v\frac{d'_v}{d_v}=\frac{m}{n}$
H = top $\sqrt{2m}$ vertices: $v\in L\Rightarrow d_v\le\sqrt{2m}$ (else $\sum_H d>2m$); $v\in H\Rightarrow d'_v\le\sqrt{2m}$. So $d_vd'_v\le\sqrt{2m}\,d_v$:

$$\mathbb{E}x^2=\tfrac1n\sum d_vd'_v\le\tfrac{2m\sqrt{2m}}{n}$$

$$\Rightarrow\ \tfrac{\text{Var}x}{(\mathbb{E}x)^2}\le\tfrac{2\sqrt2\,n}{\sqrt m}=O(\sqrt n)$$

⇒ Chebyshev: $\Theta(\sqrt n/\varepsilon^2)$ samples, 3 queries each

Unknown m need $\Theta(\frac{n}{\varepsilon^2\sqrt m})$ samples. Guess $m'=n^2,\frac{n^2}{2},\ldots$; stop when estimate $>1.5m'$. $m'>m$: est. $\le m+\varepsilon m'<1.5m'$ (no stop); $m'<m/2$: est. $\ge(1-\varepsilon)m>1.5m'$ (stop). Cost: geometric, last term dominates. Boost each round to $1-\frac{1}{\log^2n}$ (median, $O(\log\log n)$), union over $O(\log n)$ rounds ⇒ $\Theta(\frac{n\log\log n}{\varepsilon^2\sqrt m})$

#Connected comps $\pm\varepsilon n$ **(Ex. 1)** $C=\sum_v\frac{1}{|C(v)|}$ = #min-id representatives.
(a) size ≤100: BFS from sample, each discovered vertex vs all n ⇒ $\le100n$ queries; $x_i=\mathbf{1}[v_i\text{ min in comp}]$ ⇒ counting, $k=O(1/\varepsilon^2)$ ⇒ $O(n/\varepsilon^2)$
(b1) comps of size $>q$ number $\le n/q$; ignore them: $C-\frac{n}{q}\le C_{\text{small}}\le C$; $q=\frac{2}{\varepsilon}$, $\eta=\frac{\varepsilon}{2}$ ⇒ $O(k\,qn)=O(n/\varepsilon^3)$
(b2) $R\sim\mathcal{P}(q)$: $\Pr[R{=}i]=\frac1i-\frac{1}{i+1}$, $\Pr[R{\ge}j]=\frac1j$, $\mathbb{E}R=O(\log q)$; explore Rn queries; $X_v=\mathbf{1}[R\ge|C(v)|]$, $\mathbb{E}X_v=\frac{1}{|C(v)|}$ ⇒ expected $O(n\log(1/\varepsilon)/\varepsilon^2)$

6 · Lower bounds (L3)

Deterministic algorithm = decision tree; query complexity = depth. Adversary: answer 0 until the end; if $<n$ queries, an unqueried bit decides the answer (OR, XOR)

Randomized $(\mathcal{A},R)$: sample r, run deterministic A_r (randomness can always be moved to the start). Correct: $\forall x:\mathbb{E}_r[Z(x,A_r)]\ge\frac23$.

$$R(f)=\min_{(\mathcal{A},R)\text{ solves }f}\ \max_{x,r}Q(x,A_r)$$

$D_\mu(f)$ = worst-case #queries of the best *deterministic* A with $\mathbb{E}_{x\sim\mu}Z(x,A)\ge\frac23$
Yao $\forall\mu$: $D_\mu(f)\le R(f)$ (and some μ^* attains equality).
Pf: $\frac23\le\sum_x p_\mu(x)\mathbb{E}_rZ=\mathbb{E}_{x,r}Z=\sum_r p_R(r)\mathbb{E}_{x\sim\mu}Z(x,A_r)\le\max_r\mathbb{E}_{x\sim\mu}Z(x,A_r)$ ⇒ some A_r correct on μ with $\le R(f)$ queries

Template ① μ_0 (all answers 0), μ_1 (all answers 1), $I\sim\{0,1\}$, $x\sim\mu_I$ ② fix any deterministic alg. with q queries; bound its success $<\frac23$ ③ $D_\mu\ge q\Rightarrow R(f)\ge q$.
! Make the *location* of the evidence random (random position / random subset) — otherwise a deterministic alg. just looks there

TVD $\Delta_{\text{TV}}(\mu,\nu)=\frac12\sum_i|\mu(i)-\nu(i)|=\max_A|\mu(A)-\nu(A)|$. Best success at telling μ vs ν (prior $\frac12$) $=\frac12+\frac12\Delta_{\text{TV}}$ (guess the likelier; $\max(a,b)=\frac{a+b+|a-b|}{2}$). Replace ν by $\mu\Rightarrow$ failure grows $\le\Delta_{\text{TV}}$.
Lower bound: let ν_0,ν_1 be transcript distributions; show $\Delta_{\text{TV}}(\nu_0,\nu_1)<\frac13$.

Event trick: if ν_0, ν_1 are proportional to the same distribution on an event G, then $\Delta_{\mathsf{TV}} \le \max(\Pr_0[\bar{G}], \Pr_1[\bar{G}])$

OR, $R \ge n/3$ μ_1: one 1 at a uniform position. $q < \frac{n}{3}$: $p = \Pr[\text{see } 1] = \frac{1}{2}\frac{q}{n} < \frac{1}{6}$; $\Pr(I{=}0 \mid \text{all } 0) = \frac{1/2}{1/2+\frac{1}{2}(1-q/n)} < \frac{3}{5}$; success $\le p + (1-p)\frac{3}{5} = \frac{3}{5} + \frac{2}{5}p < \frac{2}{3}$. TVD view: $\Delta_{\mathsf{TV}}(\nu_0,\nu_1) = \Pr_1[\text{see } 1] < \frac{1}{3}$

XOR, $R = n$ μ uniform on $\{0,1\}^n$; any leaf with an unqueried bit ⇒ parity uniform ⇒ success $= \frac{1}{2}$

0 ones vs k ones μ_1=random k-set; $\Pr_1[\text{see } 1] \le \frac{qk}{n}$ ⇒ $R = \Theta(n/k)$; $k = \varepsilon n$ ⇒ NotOR testing needs $\Omega(1/\varepsilon)$

Which half (HW1 P2), $\Theta(n)$ UB: query first half. LB: μ uniform over n inputs; follow the all-0 path (q_0 left, q_1 right queries): correct on $\le q_0+q_1$ inputs found, plus $\le \max(\frac{n}{2}-q_0, \frac{n}{2}-q_1)$ unfound (same output) ⇒ $\le \frac{n}{2} + \max(q_0,q_1) \le \frac{n}{2} + q$ ⇒ success $\le \frac{1}{2} + \frac{q}{n}$ ⇒ $q \ge \frac{n}{6}$

Reductions solve A using an algorithm for B; each B-query costs $\le C$ A-queries ⇒ $R(B) \ge R(A)/C$.

GC: cliques on $u_{1..n}$ and $v_{1..n}$; (u_i, v_j) edge ⇔ OR bit $(i-1)n+j$; connected ⇔ OR=1; $C = 1$ ⇒ $\Omega(n^2)$.

Sorted: $a_i = i$, bit j=1 swaps pair $(2j-1, 2j)$ ⇒ $\Omega(n)$

7 · Property testing (L4, HW1, Ex. 2-3)

Def. has P → Yes; ε-far (must modify $\ge \varepsilon n$ entries / εn^2 edges) → No, each w.p. $\ge \frac{2}{3}$; otherwise either. One-sided: accept every $x \in P$ w.p. 1 (reject only with a witness)

Generic far ⇒ $\ge \varepsilon n$ bad items; sample c/ε: miss all w.p. $(1-\varepsilon)^{c/\varepsilon} \le e^{-c}$. NotOR: det. $\Theta(n)$, rand. $2/\varepsilon$

Sortedness exact: $\Omega(n)$. Attempt 1 (random adjacent pairs) fails on $7..12, 1..6$; attempt 2 (sorted random sample) fails on $2, 1, 4, 3, \ldots$ (needs $\sqrt{n}$, birthday); both miss mid-range inversions.

Binary-search tester: (distinct values) search for a_i from $j = \frac{n}{2}$; fail if an inversion vs pivot. Lemma: $i_1 \le i_2$ both succeed ⇒ at last common pivot j, $i_1 \le j \le i_2$, $a_{i_1} \le a_j \le a_{i_2}$ ⇒ successful indices sorted ⇒ change the rest ⇒ far implies $\ge \varepsilon n$ failing indices. Sample $10/\varepsilon$ ⇒ $O(\frac{\log n}{\varepsilon})$; sorted always accepted

Binary string: far ⇒ the $\frac{\varepsilon n}{2}$-th 1 precedes $\ge \frac{\varepsilon n}{2}$ zeros; sorted sample of $10/\varepsilon$ catches a red 1 and red 0 (each missed $\le e^{-5}$)

Uniformity $\Pr(x{=}y) = \sum \mu(i)^2 = \|\mu\|_2^2 \ge \frac{1}{n}$ (C-S).

Step 1: $\Delta_{\mathsf{TV}}(\mu, U) > \varepsilon$ ⇒ $\|\mu\|_2^2 \ge \frac{1+4\varepsilon^2}{n}$, since $\|\mu - U\|_2^2 = \|\mu\|_2^2 - \frac{1}{n}$ and $\|v\|_2^2 \ge \frac{1}{n}\|v\|_1^2$, $\|\mu - U\|_1 = 2\Delta_{\mathsf{TV}}$.

Step 2: $k = \frac{100\sqrt{n}}{\gamma^2}$, $\gamma = \varepsilon^2$; $Y_{ij} = \mathbf{1}[x_i{=}x_j]$, $\hat{C} = \binom{k}{2}^{-1}\sum Y_{ij}$; accept iff $\hat{C} \le \frac{1+2\varepsilon^2}{n}$.

$\mathrm{Var}(\sum Y_{ij}) \le \binom{k}{2}\|\mu\|_2^2 + 6\binom{k}{3}\|\mu\|_3^3$ (4 distinct indices: Cov 0; 3 distinct: $\le \|\mu\|_3^3 \le \|\mu\|_2^2$) ⇒ rel. var $\approx \frac{2n}{k^2} + \frac{4\sqrt{n}}{k}$ ⇒ $O(\sqrt{n}/\varepsilon^4)$; best known $\Theta(\sqrt{n}/\varepsilon^2)$

Uniformity LB $\Omega(\sqrt{n})$ **(Ex. 3)** μ_S=uniform on random S, $|S| = \frac{n}{2}$; $\Delta_{\mathsf{TV}}(\mu_S, U) = \frac{1}{2}[\frac{n}{2}\cdot\frac{1}{n} + \frac{n}{2}\cdot\frac{1}{n}] = \frac{1}{2}$. Collisions: $\delta_0 \le \binom{m}{2}\frac{1}{n}$, $\delta_1 \le \binom{m}{2}\frac{2}{n}$ (condition on S first). On D=all-distinct sequences both are uniform ($\nu_1(x)$ depends only on m) ⇒ $\Delta_{\mathsf{TV}} \le \frac{1}{2}(|\delta_1-\delta_0|+\delta_0+\delta_1) = \max\delta_i \le \frac{m(m-1)}{n}$ ⇒ success $\le \frac{1}{2} + \frac{m(m-1)}{2n}$ ⇒ $m \ge \sqrt{n/3}$

Permutation (HW1 P3) $\mathrm{dist}(a, P) = n - D$ (keep one copy per value; unchanged entries must be distinct). Far ⇒ matching of $\sum\lfloor\frac{c_v}{2}\rfloor \ge \frac{n-D}{2} \ge \frac{\varepsilon n}{2}$ equal pairs. $s = \lceil 4\sqrt{n/\varepsilon}\rceil$ distinct indices: $\Pr[I_e] = \frac{s(s-1)}{n(n-1)}$, $s(s-1) \ge \frac{A^2}{2} = \frac{8n}{\varepsilon}$ ⇒ $\mathbb{E}X \ge \frac{4n}{n-1} \ge 4$; disjoint pairs: $\mathbb{E}[I_eI_f] = \frac{s(s-1)(s-2)(s-3)}{n(n-1)(n-2)(n-3)} \le \mathbb{E}I_e\mathbb{E}I_f$ ⇒ $\mathrm{Var}X \le \mathbb{E}X$ ⇒ $\Pr[X{=}0] \le \frac{1}{\mathbb{E}X} \le \frac{1}{4}$

Bipartiteness (Ex. 2) $L = \log\frac{1}{\varepsilon}$; C: $\frac{100L}{\varepsilon}$, S: $\frac{100L}{\varepsilon^2}$; accept iff $G[C \cup S]$ bipartite; $O(L^2/\varepsilon^4)$ queries.

(2) low deg ($< \frac{\varepsilon n}{20}$) contribute $\le 0.05\varepsilon n^2$; high-deg $v \notin N(C)$ w.p. $\le (1-\frac{\varepsilon}{20})^{100L/\varepsilon} \le \varepsilon^5$, Markov on count ⇒ $\le 0.05\varepsilon n^2$ w.p. 0.99 (needs small ε).

(3) bad edge: both ends adj. to C_1 (or both to C_2); bipartite ⇒ $C_1 = C \cap A$ has none.

(4) far ⇒ every partition has $\ge 0.9\varepsilon n^2$ bad edges (else colour: adj. to C_1 → side B, else A; delete bad + outside edges $< \varepsilon n^2$).

(5) pair up S: $\Pr[\text{no bad pair}] \le (1-1.8\varepsilon)^{s/2} \le e^{-90L/\varepsilon}$; union over $2^{100L/\varepsilon}$ partitions ⇒ $e^{(L/\varepsilon)(100\ln 2 - 90)} < 0.2$

8 · Streaming (L5)

Model elements one at a time, no revisits, space ideally polylog(n)

Reservoir (1 item) replace with prob. $\frac{1}{i}$: $\Pr[x_i \text{ kept}] = \frac{1}{i}\prod_{j>i}\frac{j-1}{j} = \frac{1}{n}$; space $\log m + \log n$. k items: keep first k; for $i > k$ w.p. $\frac{k}{i}$ replace a random slot ⇒ each kept w.p. $\frac{k}{i}$ (induction: $\frac{k}{i-1}(1 - \frac{k}{i}\cdot\frac{1}{k}) = \frac{k}{i}$)

Morris increment C w.p. 2^{-C}, output $2^C - 1$. $X_i = 2^{C_i}$:

$\mathbb{E}[X_{i+1}|X_i{=}x] = 2x\cdot\frac{1}{x} + x(1-\frac{1}{x}) = x+1$ ⇒ $\mathbb{E}X_A = A+1$ (unbiased)

$\mathbb{E}[X_{i+1}^2|x] = 4x^2\frac{1}{x} + x^2(1-\frac{1}{x}) = x^2 + 3x$ ⇒ $\mathbb{E}X_i^2 = \frac{3i(i+1)}{2} + 1$

$\mathrm{Var}X_i = \frac{i(i-1)}{2} < i^2$. Mean of $\frac{10}{\varepsilon^2}$ copies: Var $< \frac{\varepsilon^2A^2}{10}$ ⇒ fail < 0.1. Space $O(\frac{\log\log n}{\varepsilon^2})$ ($C \approx \log A$)

Connectivity edge stream; keep spanning forest F (add e iff no cycle, union–find); rejected edges join already-connected vertices ⇒ F has G's components. $\le n-1$ edges, $O(n\log n)$ bits

Bipartiteness bipartite ⇔ no odd cycle. Keep forest; if $F \cup e$ closes an odd cycle (endpoints same colour by depth parity) reject. Yes ⇒ 2-colouring of F works for G: every $e \notin F$ closed an even cycle ⇒ endpoints differ

Spanner $d_G \le d_H \le k\,d_G$. Greedy: add e iff it closes no cycle of length $\le 2k$ ⇒ rejected (u,v) has $d_H(u,v) \le 2k-1$ ⇒ apply edge-by-edge on shortest paths ⇒ $(2k{-}1)$-spanner.

Size: girth $> 2k$ ⇒ first k BFS layers form a tree, each vertex $\ge d_{\min}-1$ children ⇒ $(d_{\min}-1)^k \le n$ ⇒ $d_{\min} = O(n^{1/k})$. Avg deg d ⇒ peel vertices of deg $< \frac{d}{4}$ (removes $< \frac{nd}{4} = \frac{m}{2}$ edges) ⇒ subgraph with min deg $\ge \frac{d}{4}$ ⇒ $d = O(n^{1/k})$, $m = O(n^{1+1/k})$

9 · Handy facts

- $1 + x \le e^x$ ⇒ $(1-p)^t \le e^{-pt}$; $(1-\frac{1}{k})^k < \frac{1}{e} < (1-\frac{1}{k})^{k-1}$
- e^{-1}≈.368, e^{-2}≈.135, e^{-3}≈.050, e^{-5}≈.0067, e^{-10}≈4.5×10^{-5}; $\ln 2$≈.693, $\ln 3$≈1.10
- Birthday: $\Pr[\text{collision among } m] \le \binom{m}{2}\max_i\mu(i)$ (uniform: $\frac{m^2}{2n}$); $\Theta(\sqrt{n})$ samples give a collision w.c.p.
- $(\frac{n}{k})^k \le \binom{n}{k} \le (\frac{en}{k})^k$; $\binom{n-2}{s-2}/\binom{n}{s} = \frac{s(s-1)}{n(n-1)}$; $\frac{s-a}{n-a} \le \frac{s}{n}$ for $s \le n$
- $H_n = \sum_{i\le n}\frac{1}{i} = \ln n + O(1)$; geometric sums ≈ largest term
- C-S: $(\sum v_i)^2 \le n\sum v_i^2$; $\|v\|_3^3 \le \|v\|_2^3$; $\|v\|_\infty \le \|v\|_2 \le \|v\|_1$

10 · Common pitfalls

- Chernoff needs full independence & $X_i \in [0,1]$; check δ range; without replacement / shared randomness ⇒ Chebyshev + covariance
- Yao: argue about a *deterministic* alg. with *worst-case* q queries; answers of μ_0 and μ_1 must differ
- NO instances must really be ε-far (check TVD / distance)
- State one-sided error and completeness separately; give final query/space bound
- Multiplicative approx. needs a lower bound on the target (e.g. connected ⇒ $m \ge n-1$, $a_i \ge 1$ ⇒ $S \ge n$)